

HCI Exam Study Guide: 100 Key Concepts

Based on Alan Dix et al., 3rd Edition

1. What is Human-Computer Interaction (HCI)?

Definition: HCI is the study of how people interact with computers and to what extent computers are or are not developed for successful interaction with human beings.

Core Goal: To improve the interaction between users and computers by making computers more usable and receptive to the user's needs. **Three Pillars:**

- **The Human:** How they perceive, remember, and think.
 - **The Computer:** Input/output devices, memory, and processing.
 - **The Interaction:** The dialogue and style of communication between them.
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2. The Goals of HCI: Usability

Usability is the primary goal of HCI. It is broken down into:

- **Effectiveness:** How well the system does what it's supposed to do.
 - **Efficiency:** The speed and accuracy with which users can complete tasks.
 - **Safety:** Protecting the user from dangerous conditions and undesirable situations (e.g., accidental deletion).
 - **Utility:** Providing the right kind of functionality that users need.
 - **Learnability:** How easy it is for new users to begin effective interaction.
 - **Memorability:** How easy it is to remember how to use the system.
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3. The Human: Visual Perception

Vision is the primary source of information for most computer users.

- **Brightness:** The subjective reaction to levels of light. The human eye can perceive a wide range of light levels but is sensitive to *relative* changes.
 - **Color:** Perceived through the retina's cones (sensitive to red, green, blue).
 - **Visual Acuity:** The ability to see fine detail. This decreases as we move away from the center of the visual field (the fovea).
 - **Implication:** Important information should be placed where the user is likely to be looking, and color should not be the *only* cue used (due to color blindness).
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4. The Human: Reading and Depth Perception

- **Reading:** We don't read letter by letter; we perceive words as shapes (word-shape recognition). High contrast and clear fonts improve reading speed.
 - **Depth Perception:** The brain uses cues like size, overlap, and perspective to perceive a 3D world on a 2D screen.
 - **Visual Illusions:** The brain's tendency to fill in gaps or misinterpret patterns (e.g., the Muller-Lyer illusion) can lead to design errors if not accounted for.
 - **Design Tip:** Use consistent spacing and familiar word shapes to speed up information processing.
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5. The Human: Hearing and Touch

- **Hearing:** Provides information about the environment (e.g., alerts, status). Humans can distinguish pitch, loudness, and timbre.
 - **Touch (Haptic):** Important for mobile devices and virtual reality. It provides feedback on physical interaction (e.g., button clicks, vibrations).
 - **Kinaesthetics:** The sense of position and movement of our limbs.
 - **Design Tip:** Use sound for non-critical alerts to avoid visual clutter, and use haptic feedback to confirm actions on touchscreens.
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6. The Human: Movement and Response Time

- **Response Time:** The time it takes for a person to react to a stimulus.
 - **Fitts' Law:** Predicts the time required to rapidly move to a target area.
 - *Formula:* $T = a + b \log_2(D/W + 1)$
 - D = Distance to target, W = Width of target.
 - **Design Implication:** Targets (buttons) should be large and close to the user's current cursor position to minimize movement time.
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7. The Human: Memory Systems

Humans have three types of memory:

1. **Sensory Memory:** Acts as a buffer for stimuli received through senses (iconic for vision, echoic for sound). Lasts fractions of a second.
 2. **Short-Term Memory (STM):** Also called Working Memory. Used for temporary storage of information while processing.
 3. **Long-Term Memory (LTM):** Permanent store of knowledge, facts, and experiences. **Design Tip:** Don't overload the user's working memory; provide external aids (like menus) instead of forcing recall.
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8. The Human: Short-Term Memory (STM) Capacity

- **Capacity:** Limited to approximately 7 ± 2 “chunks” of information (Miller’s Law).
 - **Chunking:** Grouping small pieces of information into larger, meaningful units (e.g., remembering a phone number as three groups of digits).
 - **Decay:** Information is lost quickly (within 10-30 seconds) unless rehearsed.
 - **Design Tip:** Group related items together and keep menu lists short (or use sub-menus) to stay within the 7 ± 2 limit.
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9. The Human: Long-Term Memory (LTM) Structure

- **Semantic Memory:** Structured knowledge of facts, concepts, and skills (e.g., knowing that a “Save” icon usually looks like a floppy disk).
 - **Episodic Memory:** Record of events and experiences in serial form (e.g., remembering the last time you used this software).
 - **Structure:** Information is stored in a network of associations.
 - **Design Tip:** Use familiar metaphors (like a “trash can” for deleting) to tap into the user’s existing semantic memory.
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10. The Human: Forgetting and Retrieval

- **Forgetting:** Information is rarely “deleted” from LTM; rather, we lose the *pathway* to it (interference).
 - **Recognition vs. Recall:** It is much easier to *recognize* something from a list (e.g., a menu) than to *recall* it from memory (e.g., a command-line code).
 - **Design Tip:** Favor **Recognition over Recall**. Use icons and menus instead of requiring users to remember complex commands.
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11. The Human: Reasoning

Humans use three types of reasoning to understand systems:

1. **Deductive:** Deriving a logical conclusion from premises (If A then B).
 2. **Inductive:** Generalizing from examples (This button worked last time, so it will work this time).
 3. **Abductive:** Reasoning from an event to its cause (The screen went black, so the battery must be dead).
- **Design Tip:** Ensure the system behaves consistently so that users' inductive and abductive reasoning leads to correct conclusions.
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12. The Human: Problem Solving

- **Problem Solving:** The process of finding a way to reach a goal.
 - **Gestalt Theory:** We perceive the “whole” rather than just the parts. Problem solving involves “insight” or restructuring the problem.
 - **Problem Space Theory:** We move through a series of “states” using “operators” to reach the goal.
 - **Design Tip:** Provide clear feedback on the current “state” of the system to help users plan their next move toward their goal.
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13. The Human: Errors (Slips vs. Mistakes)

- **Slips:** Right intention, wrong execution (e.g., meaning to click “Save” but clicking “Exit”). Usually caused by lack of attention.
 - **Mistakes:** Wrong intention (e.g., thinking “Exit” will save your work). Caused by a poor mental model of the system.
 - **Design Tip:** To prevent slips, use confirmation dialogs for destructive actions. To prevent mistakes, provide better documentation and clear system status.
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14. The Human: Emotion and Affective Computing

- **Emotion:** Affects how we use systems. Stress narrows focus (making it harder to solve complex problems), while happiness broadens it (making us more creative).
 - **Affective Computing:** Systems that can recognize and respond to human emotions.
 - **Design Tip:** An aesthetically pleasing interface (“Apple effect”) makes users more tolerant of minor usability issues because it puts them in a positive emotional state.
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15. The Computer: Input Devices (Text Entry)

- **QWERTY Keyboard:** The standard, designed to *slow down* typing to prevent mechanical jams (now just a legacy standard).
 - **Dvorak Keyboard:** Designed for speed, but failed to replace QWERTY due to habit.
 - **Chord Keyboards:** Several keys pressed at once; fast but high learning curve.
 - **Handwriting/Speech Recognition:** Natural but prone to errors (the “Ooze” vs. “Use” problem).
 - **Design Tip:** Provide keyboard shortcuts for expert users while keeping menus for novices.
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16. The Computer: Positioning and Pointing

- **Mouse:** The most common indirect pointing device. Uses a “point and click” metaphor.
 - **Touchpads/Trackballs:** Good for limited space but can be less precise.
 - **Touchscreens:** Direct manipulation. Fast but suffers from the “fat finger” problem (low precision) and “gorilla arm” (fatigue).
 - **Stylus:** High precision, good for drawing.
 - **Design Tip:** Ensure buttons are large enough for touch interaction (at least 9mm x 9mm).
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17. The Computer: Display Devices

- **CRT (Cathode Ray Tube):** Older, bulky, high power.
 - **LCD (Liquid Crystal Display):** Slim, low power, standard for laptops/monitors.
 - **OLED:** Better contrast and color, used in high-end phones.
 - **Resolution:** The number of pixels. Higher resolution = sharper images.
 - **Refresh Rate:** How often the image is redrawn. Low rates cause flicker and eye strain.
 - **Design Tip:** Use high-contrast colors (black text on white background) for better readability on all display types.
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18. The Computer: Virtual Reality and 3D Interaction

- **VR Headsets:** Provide immersion by tracking head movement.
 - **Cave Environments:** Projecting images onto walls of a room.
 - **3D Input:** Data gloves, position trackers.
 - **Challenges:** Motion sickness (lag between movement and visual update) and “presence” (the feeling of being “there”).
 - **Design Tip:** Keep latency (lag) below 20ms to prevent motion sickness in VR.
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19. The Computer: Physical Controls and Sensors

- **Dedicated Controls:** Knobs, sliders, and buttons on appliances (e.g., microwave, car dash).
 - **Sensors:** Accelerometers (detect tilt), GPS (location), Proximity (detect face).
 - **Smart-Its:** Small sensor devices that can be attached to objects to make them “smart.”
 - **Design Tip:** Physical controls should provide “affordance” (their shape should suggest how to use them—e.g., a knob suggests turning).
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20. The Computer: Memory and Processing Limits

- **RAM (Short-term):** Fast but limited. Overloading it slows down the system.
 - **Storage (Long-term):** Hard drives/SSDs. Access is slower than RAM.
 - **Processing Speed:** Affects “Interactivity.” If the computer is too slow, the user loses their “flow.”
 - **Network Latency:** Delay in data transmission.
 - **Design Tip:** Always provide a “loading” indicator (progress bar) if an action takes longer than 1.0 second.
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21. The Interaction: Models (Norman's Cycle)

Donald Norman's **7 Stages of Action**:

1. Forming the **Goal**.
 2. Forming the **Intention**.
 3. Specifying the **Action**.
 4. **Executing** the action.
 5. **Perceiving** the system state.
 6. **Interpreting** the system state.
 7. **Evaluating** the outcome.
- **Design Tip:** Design should bridge the gaps between these stages.
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22. The Interaction: Gulfs of Execution and Evaluation

- **Gulf of Execution:** The difference between the user's goals and the actions allowed by the system. (Can I do what I want?)
 - **Gulf of Evaluation:** The difference between the system's response and the user's ability to understand it. (Did it work?)
 - **Design Tip:** Use clear labels to bridge the Gulf of Execution and immediate feedback to bridge the Gulf of Evaluation.
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23. The Interaction: Abowd and Beale's Framework

An extension of Norman's model with four components:

- **User (U)**
 - **Input (I)**
 - **System (S)**
 - **Output (O)** Interaction involves translations between these components. Errors occur when a translation is “lost” (e.g., the user's intent isn't captured by the input device).
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24. The Interaction: Ergonomics

Ergonomics (or Human Factors) is the study of the physical characteristics of interaction.

- **Arrangement of controls:** Grouping by function or frequency of use.
 - **Surrounding environment:** Lighting, seating, noise.
 - **Health issues:** RSI (Repetitive Strain Injury), eye strain.
 - **Design Tip:** Critical controls (like “Emergency Stop”) should be physically distinct and easy to reach.
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25. The Interaction: Interaction Styles

- **Command Line:** Powerful, flexible, but high learning curve (recall-based).
 - **Menus:** Easy to learn, structured, but can be slow (recognition-based).
 - **Natural Language:** Familiar but ambiguous (hard for computers to parse).
 - **Question/Answer:** Simple, good for novices (e.g., installation wizards).
 - **Form-fill:** Good for data entry.
 - **Design Tip:** Use “Wizards” for complex, infrequent tasks.
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26. The Interaction: WIMP Interface

WIMP stands for:

- **Windows:** Areas of the screen for different tasks.
 - **Icons:** Small pictures representing files or actions.
 - **Menus:** Lists of options.
 - **Pointers:** Controlled by a mouse to select items.
 - **Toolbars/Palettes:** Provide quick access to frequent functions.
 - **Design Tip:** Icons should be simple and universally understood (e.g., a magnifying glass for “Search”).
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27. The Interaction: Interactivity and Feedback

- **Interactivity:** The dialogue between user and system.
 - **Feedback:** The system's response to a user's action.
 - **Types of Feedback:**
 - *Immediate:* Cursor moves.
 - *Intermediate:* Button changes color when clicked.
 - *Final:* The file is saved.
 - **Design Tip:** Never leave the user wondering if their click "registered." Provide a visual or auditory "click" immediately.
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28. Paradigms: Time-sharing and VDUs

- **Batch Processing:** Early computers; users submitted jobs and waited hours.
 - **Time-sharing:** Multiple users using one computer at once (1960s). Led to the need for interactive “Dialogue.”
 - **Video Display Units (VDU):** Allowed users to see what they were typing in real-time.
 - **Key Shift:** From “programming” to “interaction.”
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29. Paradigms: Personal Computing and WIMP

- **Personal Computing:** Computers small and cheap enough for individuals (1970s/80s).
 - **WIMP:** The Xerox Alto and later Apple Macintosh made computers accessible to non-experts through a graphical user interface (GUI).
 - **Key Shift:** Computers became tools for everyone, not just scientists.
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30. Paradigms: Metaphor and Direct Manipulation

- **Metaphor:** Using real-world concepts to explain digital ones (e.g., “Desktop,” “Folders,” “Trash”).
 - **Direct Manipulation:** Clicking and dragging objects instead of typing commands.
 - *Features:* Visibility of objects, incremental actions, and immediate feedback.
 - **Key Shift:** “What you see is what you get” (WYSIWYG).
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31. Paradigms: Hypertext and the Web

- **Hypertext:** Non-linear browsing using links (Memex, Xanadu).
 - **WWW:** Tim Berners-Lee combined hypertext with the internet.
 - **Key Shift:** Information became a global, interconnected web rather than isolated files.
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32. Paradigms: Ubiquitous Computing

- **Ubiquitous Computing (UbiComp):** Computers “disappearing” into the background (Mark Weiser).
- **Context-Awareness:** Computers that know where you are and what you are doing (e.g., a phone that silences itself in a meeting).
- **Key Shift:** Interaction happens everywhere, not just at a desk.

Part 2: Design Process

33. What is Interaction Design?

Interaction design is more than just designing a user interface; it is the practice of designing interactive digital products, environments, systems, and services. While traditional design might focus on form, interaction design focuses on **behavior**—how the system responds to the user and how the user achieves their goals.

| Feature | Traditional Design | Interaction Design |

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Focus
Outcome
User Role

34. The Design Process: Four Key Activities

The interaction design process is generally iterative and consists of four main activities that repeat until the desired usability goals are met:

1. **Establishing Requirements:** Understanding the users, their tasks, and the context of use.
 2. **Designing Alternatives:** Creating conceptual and physical designs that meet the requirements.
 3. **Prototyping:** Building interactive versions of the designs so they can be communicated and assessed.
 4. **Evaluating:** Testing the prototypes with users or experts to identify problems.
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35. User Focus and Personas

To design effectively, you must know who you are designing for. **Personas** are fictional, yet realistic, descriptions of typical users. They help the design team stay focused on user needs rather than their own preferences. A good persona includes a name, demographic data, goals, frustrations, and a brief “day in the life” story.

36. Scenarios and Storyboards

Scenarios are “stories for design.” They describe a specific user performing a specific task in a specific context. **Storyboards** are visual versions of scenarios, often using sketches to show the sequence of steps a user takes. These tools are essential for communicating the “flow” of an interaction before any code is written.

37. Navigation Design

Navigation design determines how a user moves through a system. It involves creating a **Site Map** (the structure of the information) and **Task Flows** (the paths users take to complete tasks).

***Design Rule:** Always tell the user where they are, where they can go, and how they can get back to the start.*

38. Screen Design: Layout and Grouping

The layout of a screen should reflect the structure of the task. Elements that are related should be grouped together visually using white space, borders, or similar colors. This follows the **Gestalt Principles** of proximity and similarity, helping users process information more quickly.

39. Screen Design: Color and Typography

Color should be used sparingly and consistently to draw attention or group items. Typography (fonts) must be legible; sans-serif fonts are generally preferred for screen reading.

| Element | Best Practice |

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Color
Fonts
Alignment

40. Prototyping: Low-fidelity vs. High-fidelity

Low-fidelity prototypes (like paper sketches) are fast and cheap, making them ideal for exploring early ideas. **High-fidelity prototypes** (like interactive digital mockups) look and behave like the final product, which is better for detailed user testing and gaining stakeholder approval.

41. The Software Life Cycle: Waterfall vs. Iterative

The **Waterfall Model** is a linear process (Requirements -> Design -> Code -> Test). It is risky for HCI because user feedback comes too late. The **Iterative Model** (like the Spiral Model or Agile) incorporates feedback at every stage, allowing the design to evolve based on actual user performance.

42. Usability Engineering

Usability Engineering is the process of defining specific, measurable usability goals early in the project. These goals are often expressed as **metrics**, such as “90% of users should be able to complete the checkout task in under 2 minutes without errors.”

43. Design Rationale

Design Rationale is the explicit recording of why a design decision was made. Common methods include:

- **IBIS (Issue-Based Information System):** Tracks issues, positions, and arguments.
 - **QOC (Questions, Options, Criteria):** A table that compares different design options against specific usability criteria.
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44. Design Rules: Principles of Usability

Alan Dix identifies three main categories of design principles:

1. **Learnability:** How easily a new user can master the system.
 2. **Flexibility:** The variety of ways the user and system exchange information.
 3. **Robustness:** The level of support provided to the user in achieving and assessing their goals.
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45. Learnability Principles

- **Predictability:** Can the user determine the effect of a future action based on the past?
 - **Synthesizability:** Can the user see the effect of a past action on the current state?
 - **Familiarity:** Does the system match the user's real-world knowledge?
 - **Consistency:** Does the system behave the same way in similar situations?
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46. Flexibility and Robustness Principles

- **Dialogue Initiative:** Does the user or the system control the flow? (User-led is usually better).
 - **Multi-threading:** Can the user perform more than one task at a time?
 - **Observability:** Can the user see the internal state of the system?
 - **Recoverability:** How easily can the user fix an error (e.g., Undo)?
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47. Standards and Guidelines

Standards are high-level, authoritative requirements (e.g., ISO 9241). **Guidelines** are lower-level, more specific suggestions (e.g., Apple's Human Interface Guidelines). Standards provide the "what," while guidelines provide the "how."

48. Shneiderman's 8 Golden Rules

Ben Shneiderman's classic rules for interface design:

1. Strive for **consistency**.
 2. Enable frequent users to use **shortcuts**.
 3. Offer **informative feedback**.
 4. Design dialogs to yield **closure**.
 5. Offer **error handling**.
 6. Permit easy **reversal of actions**.
 7. Support **internal locus of control**.
 8. Reduce **short-term memory load**.
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49. Nielsen's 10 Usability Heuristics

Jakob Nielsen's widely used principles for interaction design:

- Visibility of system status.
 - Match between system and the real world.
 - User control and freedom.
 - Consistency and standards.
 - Error prevention.
 - Recognition rather than recall.
 - Flexibility and efficiency of use.
 - Aesthetic and minimalist design.
 - Help users recognize, diagnose, and recover from errors.
 - Help and documentation.
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50. Implementation Support: Windowing Systems

A windowing system manages the physical screen, allowing it to be divided into multiple independent areas. It provides a “base layer” for applications to draw their interfaces and handles input from the mouse and keyboard.

51. Toolkits and Widgets

Toolkits are libraries of pre-built interface components called **widgets** (e.g., buttons, scrollbars, menus). Using toolkits ensures a consistent look and feel across different applications and saves the programmer from “reinventing the wheel.”

52. UIMS and Interface Builders

User Interface Management Systems (UIMS) separate the interface design from the application logic. **Interface Builders** (like Xcode or Visual Studio) allow designers to “drag and drop” widgets to create a GUI without writing code for the visual layout.

53. Evaluation: What, Why, and When?

Evaluation is the process of checking if a design meets user needs.

- **Formative Evaluation:** Done *during* design to identify problems and guide changes.
 - **Summative Evaluation:** Done *after* completion to measure the final quality.
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54. Evaluation: Expert Analysis (Cognitive Walkthrough)

In a **Cognitive Walkthrough**, an expert “walks through” a task step-by-step from the perspective of a new user. For each step, they ask:

1. Will the user try to achieve the right effect?
 2. Will the user notice that the correct action is available?
 3. Will the user associate the correct action with the desired effect?
 4. If the correct action is performed, will the user see that progress is being made?
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55. Heuristic Evaluation

In **Heuristic Evaluation**, a small group of experts inspects the interface to see if it follows established design rules (like Nielsen's 10 Heuristics). It is fast, cheap, and very effective at finding major usability "bugs."

56. Model-based Evaluation

Model-based evaluation uses mathematical models (like **GOMS**) to predict how long it will take a user to complete a task. This allows designers to compare different designs without actually testing them with real people.

57. Evaluation through User Participation

This involves observing real users as they use the system.

- **Think Aloud:** The user describes what they are doing and thinking as they perform a task.
 - **Co-discovery:** Two users work together, talking to each other, which provides more natural feedback.
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58. Query Techniques: Interviews and Questionnaires

- **Interviews:** Good for exploring deep user feelings and motivations.
 - **Questionnaires:** Good for gathering quantitative data (e.g., “On a scale of 1-5, how easy was this to use?”) from a large number of people.
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59. Physiological Monitoring

Advanced evaluation methods that measure the user's physical response:

- **Eye Tracking:** Shows exactly where the user is looking on the screen.
 - **EEG/Heart Rate:** Measures stress levels and emotional engagement.
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60. Controlled Experiments

A formal scientific method used to test a specific **hypothesis** (e.g., “Menu A is faster than Menu B”). It involves an **Independent Variable** (the thing you change) and a **Dependent Variable** (the thing you measure, like time or errors).

61. Universal Design: The 7 Principles

Universal design is the design of products to be usable by all people, to the greatest extent possible, without the need for adaptation.

1. Equitable Use.
 2. Flexibility in Use.
 3. Simple and Intuitive Use.
 4. Perceptible Information.
 5. Tolerance for Error.
 6. Low Physical Effort.
 7. Size and Space for Approach and Use.
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62. Multi-modal Interaction

Interaction that uses more than one human sense (sight, sound, touch). Examples include:

- **Speech Input/Output:** Good for hands-free use.
 - **Non-speech Audio:** Using “earcons” (musical icons) to provide status updates.
 - **Haptic Feedback:** Vibrations in a game controller or phone.
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63. Designing for Diversity

Designers must consider users with different abilities:

- **Visual Impairment:** Use screen readers and high-contrast modes.
 - **Hearing Impairment:** Use visual alerts and captions.
 - **Physical Disability:** Provide keyboard-only navigation and larger targets.
 - **Age:** Consider the needs of children (playful, simple) and the elderly (larger fonts, clearer navigation).
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64. User Support: Help and Documentation

Even the best-designed system needs support.

- **Quick Reference:** For expert users to look up shortcuts.
 - **Tutorial:** For new users to learn the basics.
 - **Full Explanation:** The complete manual for complex features.
 - **Context-Sensitive Help:** “Tooltips” that appear when you hover over an icon.
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65. Adaptive Help Systems

An **Adaptive Help System** monitors the user's behavior and offers help proactively. For example, if a user tries and fails to perform a task three times, the system might say, "It looks like you're trying to save a file. Would you like help?"

Part 3: Models and Theories

66. Cognitive Models: GOMS

GOMS stands for:

- **Goals:** What the user wants to achieve (e.g., “Save a file”).
 - **Operators:** The physical and mental actions the user performs (e.g., “Move mouse”).
 - **Methods:** The sequence of operators to achieve a goal (e.g., “Go to File menu -> Click Save”).
 - **Selection:** Rules for choosing between different methods (e.g., “If using keyboard, use Ctrl+S”).
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67. Cognitive Complexity Theory (CCT)

CCT is an extension of GOMS that models the complexity of a task by comparing the user's "goal structure" to the system's "rule structure." It is used to predict how long it will take to *learn* a system.

68. Production Systems

Production systems are “If-Then” rules that represent human knowledge and behavior.

Rule Example: *IF (goal is save) AND (cursor is on file) THEN (click left mouse button). These models help designers understand the mental steps a user must take to complete a task.*

69. Linguistic Models: BNF and TAG

Linguistic models view interaction as a “language” between user and computer.

- **BNF (Backus-Naur Form):** Describes the grammar of the interaction (e.g., “Command = Action + Object”).
 - **TAG (Task-Action Grammar):** Focuses on the *consistency* of the language. For example, “Copy” and “Cut” should have similar command structures.
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70. Physical and Device Models: KLM

The **Keystroke-Level Model (KLM)** is a simple way to predict task time for expert users. It assigns a time value to each action:

- **K:** Keystroke (0.2s)
 - **P:** Pointing (1.1s)
 - **H:** Homing (hands move from mouse to keyboard) (0.4s)
 - **M:** Mental preparation (1.35s)
 - **R:** Response time of the system.
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71. Cognitive Architectures

Cognitive architectures like **ACT-R** or **SOAR** are complex computer models that simulate the entire human brain—including memory, perception, and reasoning—to predict how users will interact with new interfaces.

72. Socio-Organizational Issues: Stakeholders

Stakeholders are people who are affected by a system.

- **Primary Stakeholders:** The actual users.
 - **Secondary Stakeholders:** People who receive output from the system (e.g., managers).
 - **Tertiary Stakeholders:** People who are affected by the system's success or failure (e.g., customers).
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73. Requirements Capture: Stakeholder Analysis

To design a successful system, you must understand the needs of *all* stakeholders, not just the primary users. This often involves:

- **Contextual Inquiry:** Observing people in their actual work environment.
 - **Participatory Design:** Involving users directly in the design process as co-designers.
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74. Communication and Collaboration Models

HCI is not just about one person and one computer; it is often about people working together through computers (**CSCW** - Computer-Supported Cooperative Work).

- **Synchronous:** Real-time (e.g., video call).
 - **Asynchronous:** Different times (e.g., email).
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75. Face-to-Face Communication

Face-to-face communication is the “gold standard.” It includes:

- **Verbal:** The words spoken.
 - **Non-verbal:** Eye contact, gestures, and body language.
 - **Back-channeling:** Small sounds (“uh-huh”) that show the listener is paying attention.
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76. Conversation Analysis

Conversation analysis is the study of how people take turns in a conversation. Computers often fail at this because they don't understand the subtle cues (like pauses or eye contact) that signal when a turn is ending.

77. Text-based Communication

Text-based communication (email, chat) lacks non-verbal cues. This leads to:

- **Flaming:** More aggressive behavior due to a lack of empathy.
 - **Ambiguity:** Sarcasm or humor is often misunderstood.
 - **Emoticons:** Used to replace missing facial expressions.
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78. Task Analysis: Hierarchical Task Analysis (HTA)

HTA is the process of breaking a complex task down into a hierarchy of sub-tasks and “plans” for when to perform them.

- **Task:** “Make a cup of tea.”
 - **Sub-tasks:** 1. Boil water, 2. Add tea bag, 3. Pour water.
 - **Plan:** “Do 1, then 2, then 3.”
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79. Knowledge-based Analysis

Knowledge-based analysis focuses on what the user needs to *know* to perform a task. It uses “Taxonomies” (classifications) to organize objects and actions in the system.

80. Entity-Relationship Based Techniques

These techniques focus on the “objects” (entities) in the system and how they relate to each other. For example, in a library system, entities would be “Book,” “Borrower,” and “Librarian.”

81. Dialog Notations: State Transition Networks (STN)

An **STN** uses circles (states) and arrows (transitions) to show the flow of an interaction.

- **State:** “Main Menu.”
 - **Transition:** “Click Save.”
 - **New State:** “Save Dialog.” STNs are excellent for modeling simple, linear interfaces.
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82. Petri Nets and Grammars

- **Petri Nets:** A more complex notation that can model **parallel** actions (two things happening at once).
 - **JSD (Jackson System Development):** A grammar-based notation used to model the sequence of events in a system.
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83. Models of the System: Formalisms

Formal models use mathematical notation (like **Z** or **VDM**) to define exactly how a system should behave. While hard to write, they can be used to “prove” that a system is safe or correct.

84. The PIE Model

The PIE model is a formal interaction model:

- **P:** Program (the internal state).
 - **I:** Input (what the user does).
 - **E:** Effect (what the user sees). It helps designers understand how inputs are transformed into observable effects.
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85. Modeling Rich Interaction: Status-Event Analysis

Status-event analysis distinguishes between:

- **Status:** A continuous value (e.g., the position of a mouse).
 - **Event:** A discrete occurrence (e.g., a mouse click). This is essential for modeling modern, sensor-based systems like a car's courtesy light.
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Part 4: Outside the Box

86. Groupware: Systems and CMC

Groupware is software designed to support group work.

- **CMC (Computer-Mediated Communication):** Tools for talking (email, chat).
 - **Shared Workspaces:** Tools for doing (Google Docs, shared whiteboards).
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87. Shared Artifacts and Awareness

In a shared workspace, users need **Workspace Awareness**—knowing who else is there and what they are doing. This is often achieved through:

- **Telepointers:** Showing other people's cursors.
 - **Change Tracking:** Showing who edited what.
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88. Social Issues in Groupware

Groupware often fails because of social, not technical, issues:

- **Critical Mass:** Not enough people use it.
 - **Prisoner's Dilemma:** One person does all the work while others benefit.
 - **Privacy:** Users don't want their every move tracked.
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89. Ubiquitous Computing: Applications

Ubiquitous computing (UbiComp) is about computers being everywhere.

- **Vertical:** Specialized devices (e.g., a smart fridge).
 - **Horizontal:** General devices (e.g., a smartphone).
 - **Key Challenge:** Designing for “attention”—how do we interact with a device without looking at it?
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90. Context-aware Systems

A context-aware system uses sensors to understand the user's situation:

- **Location:** Where are you?
 - **Identity:** Who are you?
 - **Activity:** What are you doing?
 - **Time:** When is it?
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91. Tangible Interfaces

Tangible interfaces allow users to interact with digital information through physical objects. For example, moving a physical block on a table to control a digital map on a screen.

92. Ambient Displays

Ambient displays provide information in the background of our environment without requiring our direct attention. For example, a lamp that changes color based on the weather forecast or the stock market.

93. Virtual Reality: Presence and Immersion

- **Immersion:** The technical capability of the system to provide a realistic experience.
 - **Presence:** The psychological feeling of “being there.”
 - **Simulator Sickness:** A common side effect of VR caused by a mismatch between the eyes and the inner ear.
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94. Augmented Reality (AR)

Unlike VR, which replaces the real world, AR overlays digital information on top of it.

- **Examples:** Pokemon Go, Google Glass, or a car's heads-up display.
 - **Key Challenge:** "Registration"—keeping the digital object perfectly aligned with the physical world.
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95. Information Visualization

Information visualization uses graphical representations to help people understand complex data.

- **Focus+Context:** Showing detailed information for one area while keeping the overall structure visible (e.g., a “fish-eye” lens).
 - **Brushing and Linking:** Selecting data in one view highlights it in all other views.
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96. Hypertext and Navigation

Hypertext is non-linear text. The biggest problem is “**Lost in Hyperspace**”—users forgetting where they are or how they got there.

- **Solution:** Use “Breadcrumbs” and “Back” buttons to provide a sense of history and location.
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97. World Wide Web: Architecture

The Web is a client-server system using:

- **HTTP:** The protocol for moving data.
 - **HTML:** The language for structuring pages.
 - **URL:** The address of a resource. The “stateless” nature of HTTP makes it hard to build complex interactive applications (like shopping carts).
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98. Web Usability Issues

Web design is different from software design because:

- **User is in control:** They can leave at any time.
 - **Navigation is key:** The user arrives at any page, not just the “Home” page.
 - **Performance:** Users are very sensitive to slow page loads.
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99. Multimedia: Combining Media

Multimedia is the combination of text, graphics, audio, and video.

- **Coherence:** The different media must work together toward a single goal.
 - **Synchronization:** Audio and video must be perfectly timed (e.g., lip-syncing).
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100. Future Trends in HCI

The future of HCI is moving toward:

- **Brain-Computer Interfaces (BCI):** Controlling computers with thoughts.
- **AI-driven Interaction:** Computers that anticipate our needs.
- **Invisible Interfaces:** Interaction through natural gestures and voice without any visible screen.
- **Ethical HCI:** Focusing on privacy, mental health, and social impact.